

NAME: \_\_\_\_\_

PERIOD: \_\_\_\_\_

DATE: \_\_\_\_\_

## What Can These Data Support?

AP Statistics · Topic 1.1 (CED 1.1) · B: 2026-09-08 · E: 2026-09-09

### [STAR] TODAY'S PROBLEM

A newspaper asked the 12 students in one AP Statistics class about sleep before a 10-point quiz; no sleep amount was assigned. The table shows the pairs.

**Claim 1:** “In this class, students who slept more tended to score higher.”

**Claim 2:** “Sleeping more raises quiz scores for all students at our school.”

Sleep and quiz scores

| Student | Sleep (h) | Quiz (/10) |
|---------|-----------|------------|
| Aiden   | 5.0       | 4          |
| Bea     | 7.0       | 7          |
| Cruz    | 8.5       | 10         |
| Dev     | 6.5       | 7          |
| Elise   | 8.0       | 8          |
| Fiona   | 7.5       | 8          |
| Gabe    | 5.5       | 5          |
| Hana    | 9.0       | 10         |
| Imani   | 7.0       | 7          |
| Jalen   | 6.0       | 6          |
| Kira    | 8.0       | 9          |
| Leo     | 4.5       | 9          |

### FIRST TAKE — before the video (5 min)

Which claim, if either, would you print? Name one detail behind your choice.

### [BOOK] CLAIMS NEED CONTEXT AND EVIDENCE (keep on the page)

A number or pattern is meaningful only in **context**: who or what was measured, what variable was recorded, when it was recorded, and its units. A statement that says *tended to* allows exceptions.

A pattern can describe the individuals measured. To generalize to a larger group, the people measured must represent that group. To claim **cause**, look for an assigned comparison in an experiment; an observed pattern alone leaves other explanations possible.

**Finish after the video:**

**DOK 1** (a) State the statistical question and identify both variables, with units.

**DOK 2** (b) Describe the relationship for these students. Cite values for the pattern and its exception.

**DOK 3** (c) Which claim is supported? Cite table evidence for Claim 1. Explain how one class and no assigned comparison limit Claim 2. Name one kind of additional data and the limit it would address.

**Frame:** Claim \_\_\_\_\_ is supported because these \_\_\_\_\_ show \_\_\_\_\_.

**Frame:** Claim 2 goes beyond the data because \_\_\_\_\_; data from \_\_\_\_\_ would address \_\_\_\_\_.

**EXIT — turn this in**

One thing the video changed about my first take: \_\_\_\_\_